

Exercises I.2

Connor Mooneyhan

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2.1. How many different bijections are there between a set S with n elements and itself?

Proof. It's the same as the number of permutations $|P_n|$ on a set with n elements. To show this, let $f : S \rightarrow S$ be a bijection and P_n the set of permutations of $\{a \in \mathbb{Z} : 1 \leq a \leq n\}$. Also, identify each $s_j \in S$ with some integer j between 1 and n (inclusive). Then for each $s_j \in S$, $f(s_j) = s_k$ for some k . We thus identify f with the permutation mapping each j to k , where k is the index of $f(s_j)$. This is clearly a permutation, and each permutation can define a bijection on S in the reverse way. $\square$

2.2. [Rewritten] Prove that f has a right-inverse if and only if it is surjective. You may assume the axiom of choice.

Proof.

($\implies$) Let $f : A \rightarrow B$ and $g : B \rightarrow A$ such that g is a right-inverse of f , so $f \circ g = id_B$. Then given $b \in B$,

$$b = id_B = (f \circ g)(b) = f(g(b)).$$

In other words, f maps $g(b)$ to b , and since the choice of b was arbitrary, f is surjective.

($\impliedby$) Let $f : A \rightarrow B$ be surjective. Define $g : B \rightarrow A$ by assigning each $b \in B$ to a choice of $x_b \in f^{-1}(b)$, where $f^{-1}(b)$ denotes the fiber of f over b as discussed in this section. In particular, $g(b) = x_b$. Observe that

$$(f \circ g)(b) = f(g(b)) = f(x_b) = b,$$

so in fact $f \circ g = id_B$, from which we can conclude that g is a right-inverse of f . $\square$

2.3. Prove that the inverse of a bijection is a bijection and that the composition of two bijections is a bijection.

Proof. Let $f : A \rightarrow B$ be a bijection. Then given $g, h : Z \rightarrow B$ where Z is an arbitrary set, we have

$$\begin{aligned} f^{-1} \circ g &= f^{-1} \circ h \\ \implies f \circ f^{-1} \circ g &= f \circ f^{-1} \circ h \\ \implies id_B \circ g &= id_B \circ h \\ \implies g &= h, \end{aligned}$$

so we conclude that f^{-1} is a monomorphism and thus is injective. We also see that given $a \in A$,

$$f^{-1}(f(a)) = (f^{-1} \circ f)(a) = id_A(a) = a,$$

so there is an element of B , namely $f(a)$, whose image under f^{-1} is a . Thus f^{-1} is surjective, and we conclude that f^{-1} is a bijection.

For the second proposition, take $f : A \rightarrow B$ and $g : B \rightarrow C$ to be bijections, where A, B, C are arbitrary sets. Then given $a_1, a_2 \in A$ such that $a_1 \neq a_2$,

$$(g \circ f)(a_1) = (g \circ f)(a_2) \iff g(f(a_1)) = g(f(a_2)). \quad (1)$$

By the injectivity of f we know that $f(a_1) \neq f(a_2)$, so by the injectivity of g we know that $g(f(a_1)) \neq g(f(a_2))$, which by (1) is equivalent to saying that $(g \circ f)(a_1) \neq (g \circ f)(a_2)$. Thus $g \circ f$ is injective.

Furthermore, given $c \in C$, we see that

$$(g \circ f)((f^{-1} \circ g^{-1})(c)) = g(f(f^{-1}(g^{-1}(c)))) = c,$$

so there exists an element of A (namely $(f^{-1} \circ g^{-1})(c)$) whose image under $g \circ f$ is c . Thus $g \circ f$ is surjective, and we conclude that the composition of bijections is a bijection. $\square$

2.4. Prove that 'isomorphism' is an equivalence relation (on any set of sets).

Proof. Given any set A , id_A is a bijection, so indeed $A \cong A$. Given another set B , if $A \cong B$, then there exists a bijection $f : A \rightarrow B$, and $f^{-1} : B \rightarrow A$ is also a bijection (by Exercise 2.3), so $B \cong A$. Introducing one more set C such that $B \cong C$ there is a bijection $g : B \rightarrow C$, so (again by 2.3) we have a bijection $g \circ f : A \rightarrow C$, showing that $A \cong C$. $\square$

2.5. Formulate a notion of *epimorphism*, in the style of the notion of *monomorphism* seen in §2.6, and prove a result analogous to Proposition 2.3, for epimorphisms and surjections.

Proof. I propose that an *epimorphism* ought to be defined as any function $f : A \rightarrow B$ where given $g, h : B \rightarrow Z$ such that $g \circ f = h \circ f$, then $g = h$, where A, B, Z are arbitrary sets. We proceed to show that

$$f \text{ is surjective} \iff [g \circ f = h \circ f \implies g = h].$$

Note that if B is empty, then there is only one function (the "empty map") from $B \rightarrow Z$ given any Z , so the proposition holds. We are thus justified in assuming that B is nonempty.

($\implies$) Let $f : A \rightarrow B$ be surjective and $g, h : B \rightarrow Z$ such that $g \circ f = h \circ f$. Then given $b \in B$, there exists $a \in A$ such that $f(a) = b$. Then

$$\begin{aligned} g(b) &= g(f(a)) \\ &= (g \circ f)(a) \\ &= (h \circ f)(a) \\ &= h(f(a)) \\ &= h(b). \end{aligned}$$

Since b is arbitrary, $g = h$.

($\impliedby$) Assume that $f : A \rightarrow B$ is not surjective. Then let $Z = \{0, 1\}$, and choose some element $\beta \in B$ such that $\beta \notin \text{im } f$. Then we can define $g, h : B \rightarrow Z$ by $g(b) = 0$ ($\forall b \in B$) and

$$h(b) = \begin{cases} 0 & b \neq \beta \\ 1 & b = \beta \end{cases}$$

so that $g \circ f = h \circ f$ since g and h agree on $\text{im } f$, but $g \neq h$. $\square$

2.6. With notation as in Example 2.4, explain how any function $f : A \rightarrow B$ determines a section of π_A .

Proof. We can define $g : A \rightarrow A \times B$ by $g(a) = (a, f(a))$ ($\forall a \in A$). Clearly, given $a \in A$,

$$(\pi_A \circ g)(a) = \pi_A(g(a)) = \pi_A((a, f(a))) = a,$$

so $\pi_A \circ g = \text{id}_A$, making g a section (i.e. right-inverse) of π_A . $\square$

2.7. Let $f : A \rightarrow B$ be any function. Prove that the graph Γ_f of f is isomorphic to A .

Proof. Define $g : A \rightarrow A \times B$ as in Exercise 2.6. Then by definition of Γ_f , $\text{im } g = \Gamma_f$. Restrict the codomain of g by defining $g' : A \rightarrow \Gamma_f$ with $g'(a) = g(a)$ ($\forall a \in A$). Clearly g' is surjective by construction, so we need only show that it is injective. Exercise 2.6 makes this clear since $\pi_A \circ g' = \text{id}_A$, so g' has a left-inverse and is therefore injective. $\square$

2.8. Describe as explicitly as you can all terms in the canonical decomposition of the function $\mathbb{R} \rightarrow \mathbb{C}$ defined by $r \mapsto e^{2\pi i r}$. (This exercise matches one assigned previously. Which one?)

Proof. For this proof, we adopt a "translation" notation for sets, where given a set S consisting of elements of some group (using $+$ for the group operation) along with x an element of that group,

$$S + x := \{s + x : s \in S\}.$$

The diagram we want to describe explicitly is the following:

$$\begin{array}{ccccccc} & & & f & & & \\ & & \nearrow & & \searrow & & \\ \mathbb{R} & \xrightarrow{\pi} & \mathbb{R}/\sim & \xrightarrow[\tilde{f}]{\sim} & \text{im } f & \xrightarrow{i} & \mathbb{C} \end{array}$$

The first definition is the standard definition of $\sim$ in this context. Namely, if $x, y \in \mathbb{R}$, then $x \sim y \iff f(x) = f(y)$. We have shown earlier in the section that this is an equivalence relation. The key insight from this is that due to the period of 2π that e^{ir} has, $e^{2\pi ir}$, has a period of 1. Thus each equivalence class is of the form $\mathbb{Z} + x$ where $x \in [0, 1)$. Formally, this means that

$$\mathbb{R}/\sim = \{Z + x : x \in [0, 1)\}.$$

Thus we can define $\pi : \mathbb{R} \rightarrow \mathbb{R}/\sim$ explicitly by $\pi(r) = Z + (r - \lfloor r \rfloor)$ ($\forall r \in \mathbb{R}$). Moving on to $\tilde{f} : \mathbb{R}/\sim \rightarrow \text{im } f$, we define $\tilde{f}(Z + x) = f(x)$ ($\forall x \in [0, 1)$). And finally, $i : \text{im } f \rightarrow \mathbb{C}$ is defined by, of course, $i(z) = z$ ($\forall z \in \mathbb{Z}$).

Due to the presence of the translations of $\mathbb{Z}$, this exercise contains a piece of Exercise 1.6! $\square$

2.9. Show that if $A' \cong A''$ and $B' \cong B''$, and further $A' \cap B' = \emptyset$ and $A'' \cap B'' = \emptyset$, then $A' \cup B' \cong A'' \cup B''$. Conclude that the operation $A \amalg B$ is well-defined up to isomorphism.

Proof. Let $f : A' \rightarrow A''$ and $g : B' \rightarrow B''$ be bijections. Then define $h : A' \cup B' \rightarrow A'' \cup B''$ by

$$h(x) = \begin{cases} f(x) & x \in A' \\ g(x) & x \in B' \end{cases}$$

This is well-defined since each x is either in A' or B' but not both, since they are disjoint. Given $y \in A'' \cup B''$, either $y \in A''$, in which case $h(f^{-1}(y)) = y$, or $y \in B''$, in which case $h(g^{-1}(y)) = y$. Thus, h is surjective.

Also, if $x_1, x_2 \in A' \cup B'$ such that $f(x_1) = f(x_2)$, then either $f(x_1)$ and $f(x_2)$ are both in A'' or they are both in B'' . Without loss of generality, assume they are both in A'' . Then since f is injective, $x_1 = x_2$.

Thus h is a bijection, by which we can conclude that all ways in which "copies" of sets may be taken to achieve a disjoint union will yield isomorphic results. $\square$

2.10. Show that if A and B are finite sets, then $|B^A| = |B|^{|A|}$.

Proof. We use induction on $|A|$. If $|A| = 0$, then there exists exactly one map, the empty map, from A to any set. This coincides with the proposition if we assume $0^0 = 1$ (in case $|B| = 0$), which we shall do for convenience. If $|A| = 1$, then each $f : A \rightarrow B$ consists of a choice of $b \in B$ to which to send the $a \in A$. Thus there are $|B| = |B|^{|A|}$ such functions.

Assume, then, given A such that $|A| > 1$, the proposition holds for all sets of size less than $|A|$. Then choose one element $\alpha \in A$. Each $f : A \rightarrow B$ can be written as

$$f(a) = \begin{cases} f|_{\{\alpha\}}(a) & a = \alpha \\ f|_{A \setminus \{\alpha\}}(a) & a \neq \alpha \end{cases}$$

We can thus define a map $g : B^A \rightarrow B^{\{\alpha\}} \times B^{A \setminus \{\alpha\}}$ by

$$g(h) = (h|_{\{\alpha\}}, h|_{A \setminus \{\alpha\}}).$$

This is clearly a bijection, so

$$\begin{aligned} |B^A| &= |B^{\{\alpha\}} \times B^{A \setminus \{\alpha\}}| \\ &= |B^{\{\alpha\}}| |B^{A \setminus \{\alpha\}}| \\ &= |B|^{| \{\alpha\} |} |B|^{|A \setminus \{\alpha\}|} \\ &= |B|^1 |B|^{|A|-1} \\ &= |B|^{|A|}. \end{aligned}$$

□

2.11. In view of Exercise 2.10, it is not unreasonable to use 2^A to denote the set of functions from an arbitrary set A to a set with 2 elements (say $\{0, 1\}$). Prove that there is a bijection between 2^A and the *power set* of A .

Proof. Denote the power set of A by $\mathcal{P}(A)$. Then define $f : 2^A \rightarrow \mathcal{P}(A)$ by

$$\phi \mapsto \{x \in A : \phi(x) = 1\}$$

for all $\phi \in 2^A$.

Given $S \in \mathcal{P}(A)$, we have $f(\phi) = S$ where ϕ is the function

$$\phi(x) = \begin{cases} 1 & x \in S \\ 0 & x \notin S \end{cases}$$

for all $x \in A$. Thus f is surjective.

On the other hand, if $\phi_1, \phi_2 \in 2^A$ and $\phi_1 \neq \phi_2$, then take some $x \in A$ for which $\phi_1(x) \neq \phi_2(x)$, and by definition $x \in f(\phi_1)$ and $x \notin f(\phi_2)$ or vice versa. In either case, we see that $f(\phi_1) \neq f(\phi_2)$, so f is injective.

We conclude that 2^A and $\mathcal{P}(A)$ are isomorphic as sets, so using 2^A to denote the power set of A is justified. □